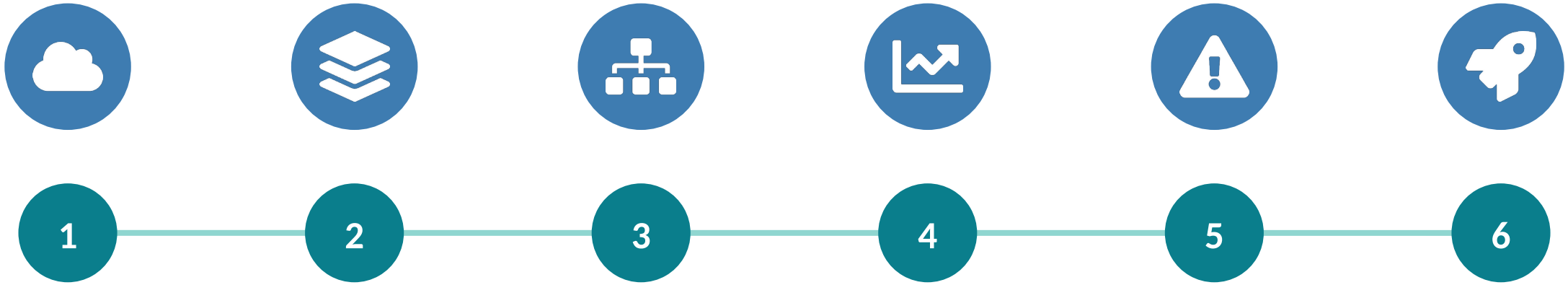
Cloud Architecture: Components, Benefits and Challenges

Building scalable, resilient, and cost-efficient systems

Presented By : Adesh Gautam Lokhande
Class : MCA 2nd Years
Section : A
Roll No. : MC250005CA
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Agenda



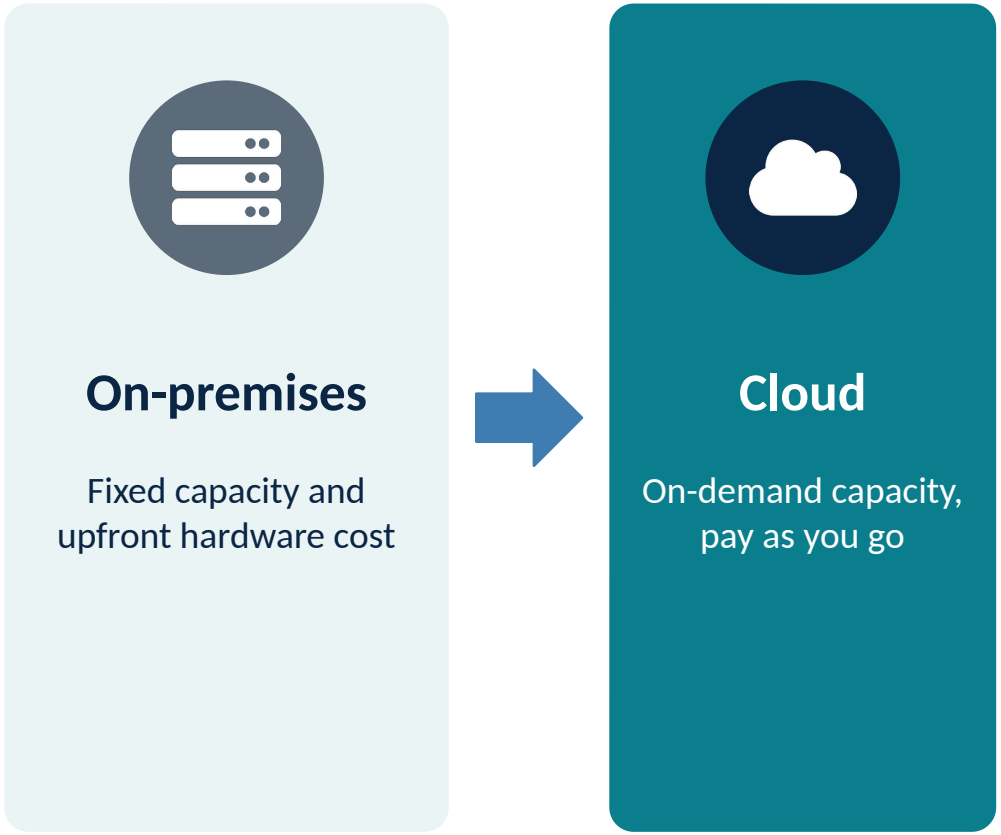
1. What is cloud architecture?
2. Core components
3. Service and deployment models
4. Key benefits
5. Challenges and risks
6. Best practices and the future

What Is Cloud Architecture?

The blueprint for how cloud components fit together to deliver applications and services

Combines compute, storage, networking and software, delivered over the internet

Replaces or extends on-premises infrastructure with on-demand resources



Core Components Overview



Front end

Browsers, apps and devices that users interact with



Network

The internet or private connections linking everything



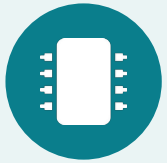
Back end

Servers, storage, databases and application logic

Cloud delivery model

IaaS | PaaS | SaaS

Key Building Blocks



Compute

Virtual machines, containers and serverless functions



Storage

Object, block and file storage



Databases

Relational (SQL) and NoSQL



Networking

VPCs, load balancers, CDNs and DNS



Security and identity

IAM, encryption and firewalls



Management and monitoring

Logging, automation and orchestration

Service Models

	IaaS e.g. AWS EC2	PaaS e.g. Google App Engine	SaaS e.g. Microsoft 365
Applications	**You manage**	**You manage**	Provider
Data	**You manage**	**You manage**	Provider
Runtime	**You manage**	Provider	Provider
Operating system	**You manage**	Provider	Provider
Virtualization	Provider	Provider	Provider
Servers	Provider	Provider	Provider
Storage and network	Provider	Provider	Provider

Managed by you Managed by the provider

IaaS: raw infrastructure, maximum control. PaaS: managed platform for developers. SaaS: ready-to-use software.

Deployment Models



Public cloud

Shared infrastructure run by a provider

Cost-effective and elastic



Private cloud

Dedicated infrastructure for one organization

Higher control and compliance



Hybrid cloud

A mix of public and private environments

Sensitive data stays private, scale in public

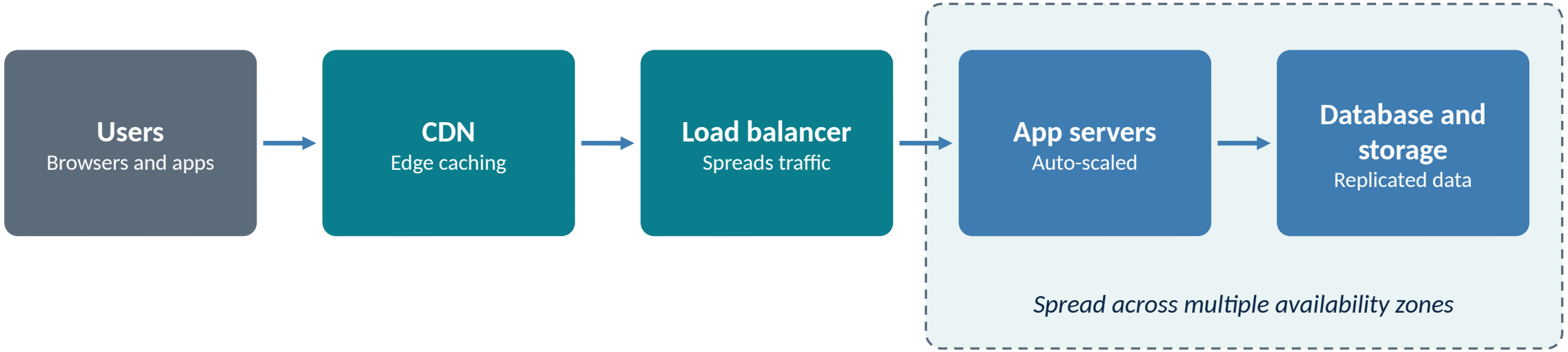


Multi-cloud

Services from multiple providers

Avoids lock-in and improves resilience

A Typical Cloud Architecture



Supporting layers

- Authentication
- Caching
- Monitoring
- Backups

Benefits: Flexibility and Cost

Scalability

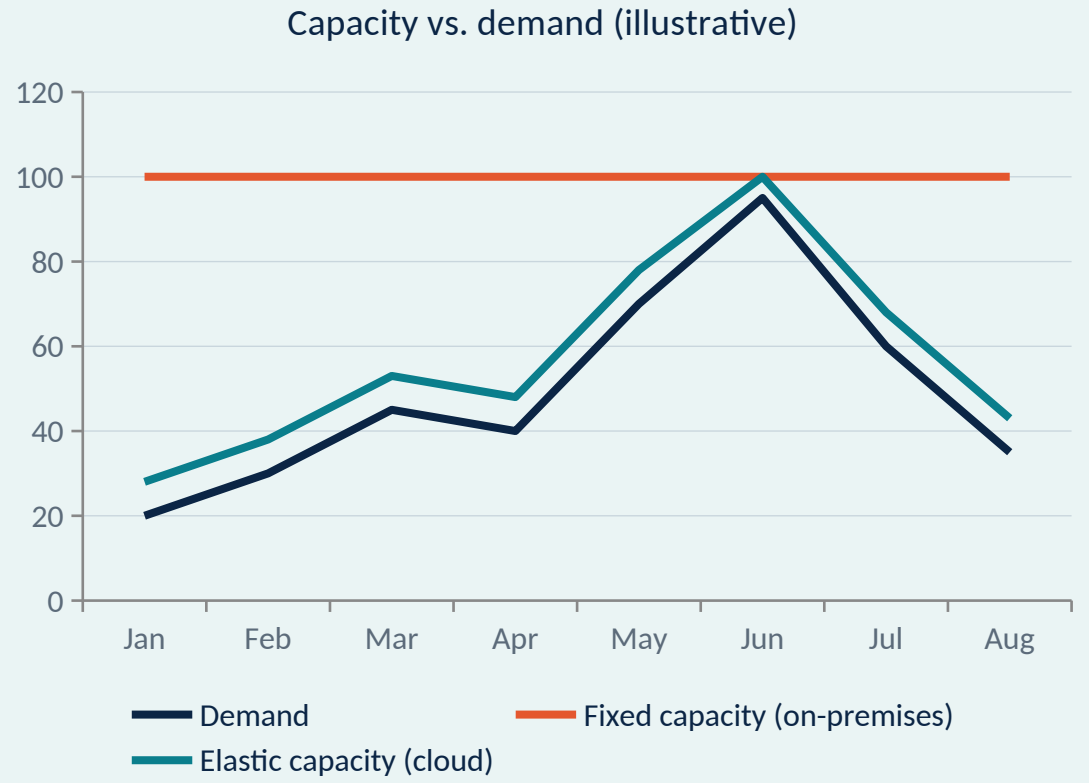
Scale up or down on demand

Cost efficiency

Pay-as-you-go with less upfront hardware spend

Speed and agility

Deploy in minutes, not weeks



Illustrative data to show the concept of elastic capacity

Benefits: Reliability and Reach



High availability

Redundancy across regions and zones keeps services running



Disaster recovery

Automated backups and failover restore service quickly



Global reach

Deploy close to users worldwide for lower latency



Innovation

Easy access to AI, analytics and IoT services

Challenges: Security and Compliance

Provider: secures the cloud

Hardware, facilities and core network

You: secure what's in the cloud

Data, identities, configuration and access



Security risks

- Misconfigurations
- Data breaches
- Insider threats



Compliance

- GDPR
- HIPAA
- Data-residency requirements

Challenges: Operations and Cost

Problem	Impact
Cost overruns	Unmonitored usage and idle resources inflate the bill
Vendor lock-in	Migrating between providers becomes difficult and costly
Downtime and dependency	Reliance on provider uptime and connectivity
Skills gap	Teams need cloud expertise to design and operate safely
Complexity	Distributed and multi-cloud environments are harder to manage



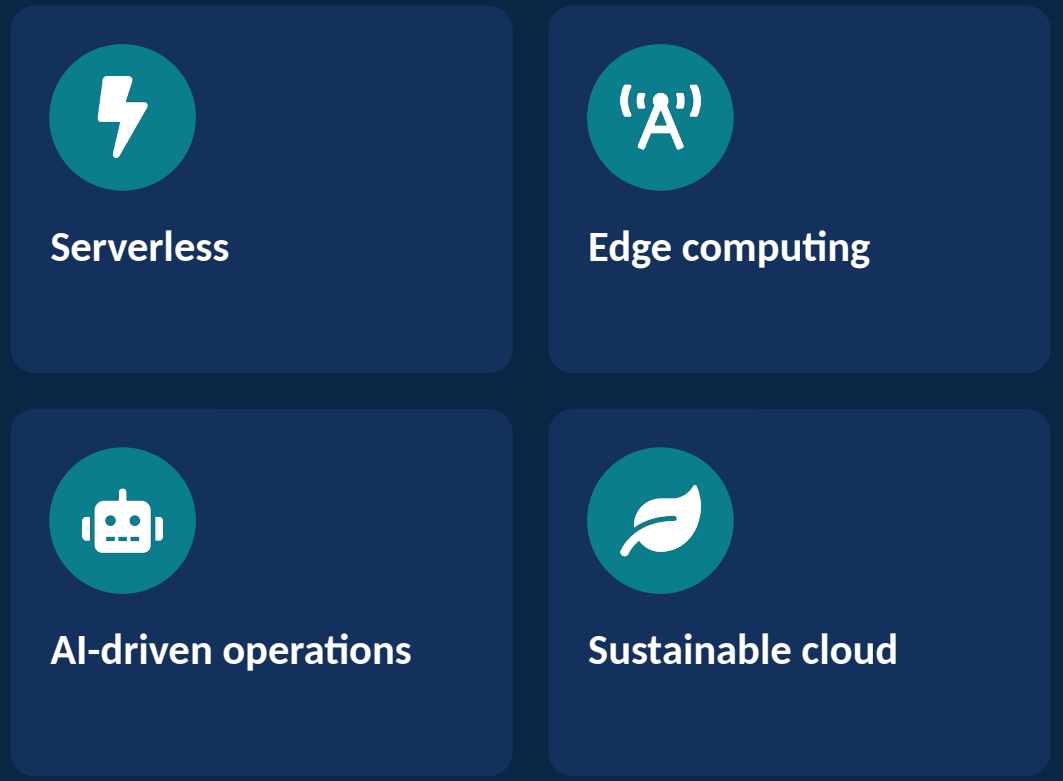
Each challenge has a proven mitigation, covered in the next slide

Best Practices

Design for failure	Use multiple zones and automated recovery
Secure by default	Apply least-privilege access; encrypt data at rest and in transit
Control costs	Use budgets, tagging and right-sizing
Automate	Infrastructure as Code with Terraform or CloudFormation
Follow a framework	Such as the AWS Well-Architected Framework

Conclusion and Future Trends

- Cloud architecture enables scalable, resilient and agile systems
- Success depends on balancing benefits against security, cost and complexity
- Design for failure, secure by default, and control costs



“Plan well, design for resilience, and optimize continuously.”

THANK YOU